

Multilinear Averages on Curves

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Set-up.

We define *multilinear averages on curves*:

$$\mathcal{M}(f_1, \dots, f_k) := \int_{\mathbb{R}^n} \prod_{j=1}^k f_j \circ \pi_j(x) a(x) dx;$$

- * a is a cutoff function;
- * each $\pi_j : \mathbb{R}^n \rightarrow \mathbb{R}^{n-1}$ is a smooth submersion.

Question: For which $p = (p_1, \dots, p_k)$ does the following estimate hold:

$$|\mathcal{M}(f_1, \dots, f_k)| \stackrel{?}{\leq} C \prod_{j=1}^k \|f_j\|_{L^{p_j}(\mathbb{R}^{n-1})}, \quad (\star)$$

where C may depend on the π_j, p_j , and a , but not on the f_j .

Main result.

Theorem (A., 2026+). Let $k \geq 3$ and $n \geq k + 1$. Then the sharp L^p estimates for $(\star)$ hold if the π_j are “polynomial-like.”

Definition: $X_j \neq 0$ are vector fields with $d\pi_j(X_j) = 0$. Then the π_j are *polynomial-like* if the X_j generate a nilpotent Lie algebra and the elements have polynomial flows.

Set-up and Notation.

- * $\pi_j^{-1}(X_j)$ is a curve.
- * The *incidence set associated to open sets $E_1, \dots, E_k$* is $\Omega = \pi_1^{-1}(E_1) \cap \dots \cap \pi_k^{-1}(E_k) \cap \{a \sim 1\}$.
- * The *average “number” of incidences per curve* is $\alpha_j := |\Omega| / |E_j|$.
- * $A \lesssim B$ means there exists $C > 0$ so that $A \leq CB$.
- * For $0 < \varepsilon \ll 1$, $A \lesssim_\varepsilon B$ means there exists $C > 0$ so that $A \leq C\varepsilon^{-C}B$.

Prior results.

All are sharp.

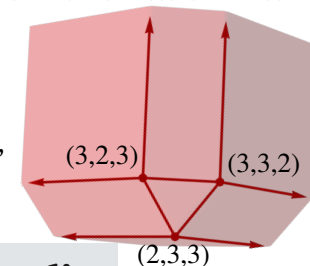
Reference	[4]	[3]	[2]	[1]
k	2	≥ 3	≥ 2	2
Endpoints?	No	No	Yes	Yes
Polynomial-like curves?	No	No	Yes	Yes
Notes			$f_j = \chi_{E_j}$	

Example.

For

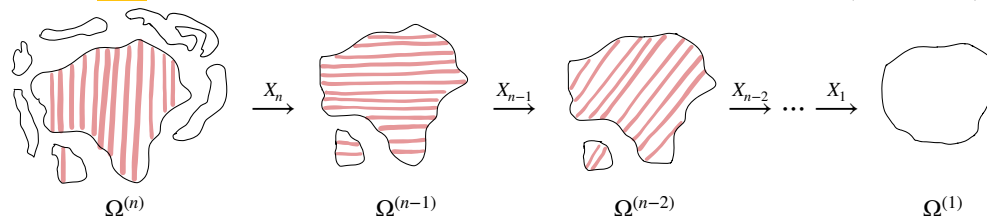
$$\mathcal{M}(f_1, f_2, f_3) = \int_{\mathbb{R}^5} f_1(x_1 - t, x_2 - t^2/2, x_3, x_4) f_2(x_1, x_2, x_3 - t, x_4 - t^2/2) f_3(x) a(t, x) d(t, x),$$

the inequality $(\star)$ is sharp for all exponents p in the polyhedron.

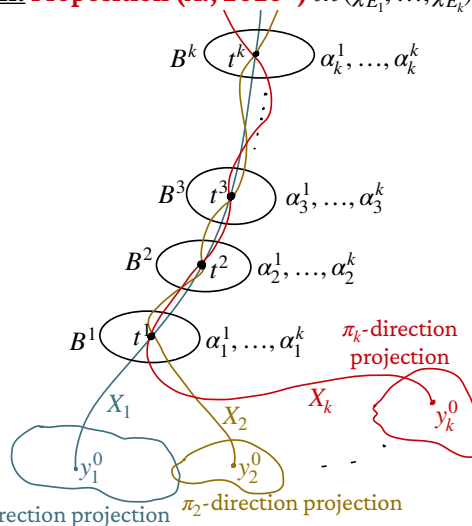


Quasi-extremal case of the restricted weak-type inequality.

Step I: (Re-)prove $(\star)$ with $f_j = \chi_{E_j}$. **Equivalently:** $|\Omega| \gtrsim \alpha_1^{b_1} \dots \alpha_k^{b_k}$, where $b_j = p_j^{-1} / (\sum p_j^{-1} - 1)$.



Step II: Proposition (A., 2026+) $\mathcal{M}(\chi_{E_1}, \dots, \chi_{E_k}) \approx \alpha_1^{b_1} \dots \alpha_k^{b_k} \implies \pi_j^{-1}(x) \cap \Omega \approx$ an interval containing “most x .”



The strong-type inequality $(\star)$.

Step III: Decompose general $f_j \rightsquigarrow \sum_{\ell} 2^{\ell} \chi_{E_{j,\ell}}$.

- * Big pieces: $|\Omega_{\ell_1, \dots, \ell_k}| \approx \alpha_{1,\ell_1}^{b_1} \dots \alpha_{k,\ell_k}^{b_k}$.
- * Bad case: ℓ_1 (e.g.) in many such Ω .
- * **New:** Proving that the ‘bad case’ cannot happen in the multilinear setting.
- * The sets $E_1, \dots, E_k$ saturate the restricted weak-type inequality $\implies$ each may be well-approximated by projections $\pi_1, \dots, \pi_k$ of Carnot-Carathéodory balls.
- * Problem: Many incidence sets project to the same $E_{j,\ell}$.
- * **New:** Such projections do not overlap substantially.

Acknowledgements and References.

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[1] Michael Christ, Spyridon Dendrinos, Betsy Stovall, and Brian Street. Endpoint Lebesgue estimates for weighted averages on polynomial curves. *American Journal of Mathematics*, 142(6):1661–1731, 2020.

[2] Philip T. Gressman. L^p -improving properties of averages on polynomial curves and related integral estimates. *Mathematical Research Letters*, 16(6):971–989, 2009.

[3] Betsy Stovall. L^p improving multilinear Radon-like transforms. *Revista Matemática Iberoamericana*, 27(3):1059–1085, December 2011.

[4] Terence Tao and James Wright. L^p improving bounds for averages along curves. *Journal of the American Mathematical Society*, 16(3):605–638, 2003.